

Sequences and progressions

[Cambridge revision]

Two formulas for the term, two for the sum — and one question that decides which pair.

LESSON 99–100

2 × 40 MIN

REVISION OF GRADE 9, §§18–21

P1 7.1–7.3

LESSON CLOCK

15'

25'

25'

25'

15'

5

Sequences and recurrences	15 min
Arithmetic progressions	25 min
Geometric progressions	25 min
Sums	25 min
Choosing the right pair	15 min
Homework	5 min

LEARNING OBJECTIVES

- Define a sequence by a formula for the n th term and by a recurrence.
- Identify an arithmetic or a geometric progression from its terms.
- Use the n th-term and sum formulas for both progressions.
- Solve problems where the number of terms is the unknown.

TEXTBOOK

National Algebra 9, pp. 190–214

Cambridge Pure Mathematics 1, pp. 148–162

Workbook Exercise 7A–7C

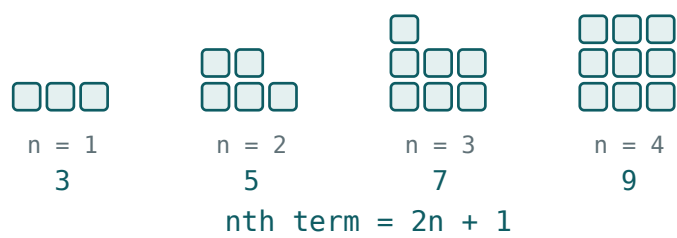
01

Explanation

Sequences and recurrences

A sequence can be given in two ways: by a **formula for the n th term**, which lets you jump straight to u_{100} , or by a **recurrence**, which tells you how to get from one term to the next.

SEQUENCE	NTH TERM	RECURRENCE
3, 7, 11, 15, ...	$u_n = 4n - 1$	$u_{n+1} = u_n + 4, u_1 = 3$
2, 6, 18, 54, ...	$u_n = 2 \cdot 3^{n-1}$	$u_{n+1} = 3u_n, u_1 = 2$
1, 4, 9, 16, ...	$u_n = n^2$	$u_{n+1} = u_n + 2n + 1$
1, 1, 2, 3, 5, ...	no simple formula	$u_{n+2} = u_{n+1} + u_n$



An arithmetic sequence plotted — the points lie on a straight line, and d is its gradient.

SIGMA NOTATION

$$\sum_{k=1}^n u_k = u_1 + u_2 + \dots + u_n$$

The letter under the sigma is a dummy: $\sum k^2$ and $\sum j^2$ are the same number.

Arithmetic progressions

Each term exceeds the one before by a fixed d .

$$u_n = a + (n - 1)d$$

The $n - 1$ matters: you take d one fewer times than the number of terms, because the first term is already there.

PROGRESSION	A	D	U_{10}
3, 7, 11, ...	3	4	39
20, 17, 14, ...	20	-3	-7
$\frac{1}{2}, 1, \frac{3}{2}, \dots$	$\frac{1}{2}$	$\frac{1}{2}$	5

TWO TERMS GIVE A AND D

From $u_4 = 11$ and $u_9 = 26$: subtracting gives $5d = 15$, so $d = 3$ and $a = 11 - 3(3) = 2$. Always subtract the two equations — never solve them one at a time.

Geometric progressions

Each term is a fixed multiple r of the one before.

$$u_n = a r^{(n-1)}$$

PROGRESSION	A	R	U_6
2, 6, 18, ...	2	3	486
16, 8, 4, ...	16	$\frac{1}{2}$	$\frac{1}{2}$
1, -2, 4, -8, ...	1	-2	-32

A NEGATIVE RATIO ALTERNATES IN SIGN

With $r = -2$ the terms swing above and below zero. Any question mentioning “alternating” or giving terms of opposite signs is telling you $r < 0$.

Two terms give a and r by dividing, not subtracting: from $u_3 = 12$ and $u_6 = 96$, $r^3 = 8$ so $r = 2$ and $a = 3$.

Sums

$$S_n = \frac{n}{2}(2a + (n-1)d) = \frac{n}{2}(a + l)$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

The second form of the arithmetic sum — *half the number of terms times the sum of the first and last* — is the fastest when the last term is known. The two geometric forms are identical; use whichever avoids negative signs.

QUESTION	USE	ANSWER
$3 + 7 + 11 + \dots$ (20 terms)	$\frac{n}{2}(2a + (n-1)d)$	820
$1 + 2 + 3 + \dots + 100$	$\frac{n}{2}(a + l)$	5050
$2 + 6 + 18 + \dots$ (8 terms)	$\frac{a(r^n - 1)}{r - 1}$	6560

Choosing the right pair

ONE TEST SETTLES IT

Take three consecutive terms. If the **differences** are equal it is arithmetic; if the **ratios** are equal it is geometric. If neither, it is neither — and the question wants something else.

When the unknown is n , the formula becomes an equation to solve — linear or quadratic for an AP, exponential for a GP.

QUESTION	EQUATION	ANSWER
how many terms of 3, 7, 11, ... sum to 820?	$\frac{n}{2}(6 + 4(n-1)) = 820$	$n = 20$
which term of 5, 8, 11, ... equals 98?	$5 + 3(n-1) = 98$	$n = 32$
which term of 3, 6, 12, ... first exceeds 1000?	$3 \cdot 2^{n-1} > 1000$	$n = 10$

N MUST BE A POSITIVE WHOLE NUMBER

A quadratic in n usually has one root that is negative or fractional. Reject it with a sentence, do not leave it in the answer.

02 Worked examples

EXAMPLE 1 An AP has $u_4 = 11$ and $u_9 = 26$. Find a , d and S_{20} .

$$① \quad a + 3d = 11, \quad a + 8d = 26$$

$$② \quad 5d = 15 \Rightarrow d = 3$$

Subtract.

$$③ \quad a = 2$$

$$④ \quad S_{20} = 10(4 + 57) = 610$$

ANSWER

$$a = 2, d = 3, S_{20} = 610$$

EXAMPLE 2 A GP has $u_3 = 12$ and $u_6 = 96$. Find a , r and S_8 .

$$① \quad \frac{ar^5}{ar^2} = \frac{96}{12} = 8$$

Divide.

$$② \quad r^3 = 8 \Rightarrow r = 2$$

$$③ \quad ar^2 = 12 \Rightarrow a = 3$$

$$④ \quad S_8 = \frac{3(2^8 - 1)}{1} = 765$$

ANSWER

$$a = 3, r = 2, S_8 = 765$$

EXAMPLE 3 How many terms of $3 + 7 + 11 + \dots$ are needed to reach a sum of 820?

$$① \quad \frac{n}{2}(6 + 4(n - 1)) = 820$$

$$② \quad n(4n + 2) = 1640$$

$$③ \quad 2n^2 + n - 820 = 0$$

$$④ \quad n = 20 \text{ (the root } -20.5 \text{ is rejected).}$$

ANSWER

$$n = 20$$

EXAMPLE 4 The first three terms of a GP are x , $x + 4$, $3x$. Find x .

$$① \quad \frac{x + 4}{x} = \frac{3x}{x + 4}$$

Equal ratios.

$$② \quad (x + 4)^2 = 3x^2$$

$$③ \quad 2x^2 - 8x - 16 = 0 \Rightarrow x^2 - 4x - 8 = 0$$

$$④ \quad x = 2 \pm 2\sqrt{3}$$

ANSWER

$$x = 2 + 2\sqrt{3} \text{ or } x = 2 - 2\sqrt{3}$$

03 Interactive model

Write two mystery sequences on the board and have the class decide AP, GP or neither by differences and ratios alone.

Arithmetic, geometric, or neither?

3, 7, 11, 15 is:

AP

GP

neither

both

2, 6, 18, 54 is:

AP

GP

neither

both

1, 4, 9, 16 is:

AP

GP

neither

both

u_{10} of 3, 7, 11, ...:

39

43

40

36

u_6 of 16, 8, 4, ...:

1

$\frac{1}{2}$

2

$\frac{1}{4}$

$1 + 2 + \dots + 100$ is:

5000

5050

10100

4950

To find r from two terms you:

subtract

divide

add

square

A negative r makes the terms:

grow

shrink

alternate in sign

constant

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Sequence	Ketma-ketlik	Последовательность
Term	Had	Член
Recurrence relation	Rekurrent munosabat	Рекуррентное соотношение
Arithmetic progression	Arifmetik progressiya	Арифметическая прогрессия
Geometric progression	Geometrik progressiya	Геометрическая прогрессия
Common difference	Ayirma	Разность
Common ratio	Maxraj	Знаменатель
Sum of n terms	n ta hadning yig'indisi	Сумма n членов
Sigma notation	Sigma belgisi	Знак суммы
Convergent	Yaqinlashuvchi	Сходящаяся

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself u_n of an AP is:

$a + nd$

$a + (n - 1)d$

ar^n

$a + (n + 1)d$

 S_n of an AP is:

$\frac{n}{2}(a + l)$

$n(a + l)$

na

$\frac{a}{1 - r}$

 u_n of a GP is:

ar^n

$ar^{(n-1)}$

$a + r^n$

anr

The GP sum formula fails when:

$$r = 0$$

$$r = 1$$

$$r < 0$$

$$a = 0$$

Three terms are geometric when:

the differences are equal

the ratios are equal

they are positive

the middle is the mean

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. u_{10} of 3, 7, 11, ...

ANSWER 39

2. u_{15} of 20, 17, 14, ...

ANSWER -22

3. u_5 of 2, 6, 18, ...

ANSWER 162

4. u_6 of 16, 8, 4, ...

ANSWER $\frac{1}{2}$

5. S_{10} of 1, 2, 3, ...

ANSWER 55

6. S_5 of 3, 6, 12, ...

ANSWER 93

7. Is 5, 10, 20, 40 an AP or a GP?

ANSWER GP, $r = 2$

Medium · the standard question

7 PROBLEMS

1. AP with $u_4 = 11$, $u_9 = 26$: find a and d

ANSWER $a = 2$, $d = 3$

2. Same AP: S_{20}

ANSWER 610

3. GP with $u_3 = 12$, $u_6 = 96$: find a and r

ANSWER $a = 3$, $r = 2$

4. Same GP: S_8

ANSWER 765

5. Which term of 5, 8, 11, ... equals 98?

ANSWER $n = 32$

6. $1 + 2 + 3 + \dots + 100$

ANSWER 5050

7. Sum of the first 20 terms of 3, 7, 11, ...

ANSWER 820

Hard · stretch and reason

7 PROBLEMS

1. How many terms of $3 + 7 + 11 + \dots$ give 820?

ANSWER $n = 20$

2. First three terms of a GP are $x, x + 4, 3x$: find x

ANSWER $2 \pm 2\sqrt{3}$

3. Which term of $3, 6, 12, \dots$ first exceeds 1000?

ANSWER $n = 10$

4. An AP has $S_{10} = 155$ and $S_{20} = 610$: find a and d

ANSWER $a = 2, d = 3$

5. The sum of the first n odd numbers

ANSWER n^2

6. A GP has $u_2 = 6$ and $u_5 = -48$: find r

ANSWER $r = -2$

7. Three numbers in AP have sum 15 and product 80: find them

ANSWER 2, 5, 8 (or reversed)

07 Homework

Homework — 5 tasks

Write a and d (or r) on their own line before every calculation.

1. An AP has $u_5 = 17$ and $u_{12} = 45$. Find a, d and S_{25} .
2. A GP has $u_2 = 10$ and $u_5 = 80$. Find a, r and S_7 .
3. How many terms of $2 + 5 + 8 + \dots$ are needed for a sum over 1000?
4. The numbers $x - 2, x, x + 6$ form a GP. Find x .

5. Show that the sum of the first n even numbers is $n(n + 1)$, and check it for $n = 6$.